

Smart Route Management and Scheduling System (SRMSS)

Project Overview

The Smart Route Management and Scheduling System (SRMSS) is a web-based application developed to digitalize and streamline public transport depot operations. The system replaces manual record keeping and spreadsheet-based scheduling with a centralized platform for route management, vehicle allocation, driver assignment, timetable scheduling, fuel monitoring, maintenance tracking, and operational reporting.

Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript (ES6)
- Bootstrap 5

Backend

- FastAPI
- Python 3.12

Database & Authentication

- Firebase Realtime Database
- Firebase Authentication
- Pyrebase4

Additional Libraries

- Jinja2 Templates
 - Pydantic
 - ReportLab (PDF Generation)
 - Google Maps API / Leaflet.js
 - Uvicorn
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Software Development Methodology

Agile Scrum

The project follows Agile Scrum methodology.

Sprint 1

- Requirement Analysis
- System Design
- UML Diagrams
- Database Design

Sprint 2

- Authentication Module
- User Management Module
- Route Management Module

Sprint 3

- Vehicle Management Module
- Driver Management Module
- Schedule Management Module

Sprint 4

- Dashboard Module
- Fuel & Maintenance Module
- Reporting Module

Sprint 5

- Testing
- Bug Fixing
- Documentation
- Deployment

System Modules

Module 1: Authentication & User Management

Features

- Login

- Logout
- Role Based Access Control
- User Registration
- User Update
- Session Management

Firestore Collections

users

```
{
  "uid": {
    "name": "",
    "email": "",
    "role": "admin"
  }
}
```

Module 2: Route Planning & Management

Features

- Create Route
- Update Route
- Delete Route
- View Route
- Route Visualization

Firestore Collection

routes

```
{
  "route_id": {
    "start_point": "",
    "end_point": "",
    "distance": "",
    "service_type": "",
    "stops": []
  }
}
```

Module 3: Vehicle Management

Features

- Add Vehicle
- Update Vehicle
- Delete Vehicle
- View Vehicle

Firestore Collection

vehicles

```
{
  "vehicle_id": {
    "registration_no": "",
    "capacity": 0,
    "mileage": 0,
    "status": ""
  }
}
```

Module 4: Driver Management

Features

- Add Driver
- Update Driver
- View Driver
- Assignment History

Firestore Collection

drivers

```
{
  "driver_id": {
    "name": "",
    "license_no": "",
    "working_hours": 0
  }
}
```

Module 5: Schedule Management

Features

- Create Schedule
- Update Schedule
- View Schedule
- Conflict Detection
- Emergency Schedule Updates

Firestore Collection

schedules

```
{
  "schedule_id": {
    "route_id": "",
    "vehicle_id": "",
    "driver_id": "",
    "departure_time": "",
    "arrival_time": ""
  }
}
```

Module 6: Depot Operations Dashboard

Features

- Active Trips
- Available Buses
- Assigned Drivers
- Vehicle Utilization
- Live Bus Tracking

Dashboard Widgets

- Total Routes
 - Active Routes
 - Total Drivers
 - Total Vehicles
 - Delayed Trips
 - Completed Trips
-

Module 7: Fuel & Maintenance Management

Features

- Fuel Consumption Logs
- Maintenance Logs
- Vehicle Service Records

Firestore Collections

fuel_logs

```
{
  "fuel_id": {
    "vehicle_id": "",
    "liters": 0,
    "date": ""
  }
}
```

maintenance_logs

```
{
  "maintenance_id": {
    "vehicle_id": "",
    "service_type": "",
    "date": ""
  }
}
```

Module 8: Reporting & Analytics

Features

- Weekly Reports
- Monthly Reports
- Route Performance Analysis
- Fuel Consumption Analysis
- PDF Export

Reports

1. Route Performance Report

2. Vehicle Utilization Report
 3. Fuel Consumption Report
 4. Maintenance Report
 5. Trip Completion Report
-

Project Structure

```
SRMSS/
|
├── app/
|   ├── main.py
|   ├── firebase_config.py
|   |
|   ├── routes/
|   |   ├── auth_routes.py
|   |   ├── route_management_routes.py
|   |   ├── vehicle_routes.py
|   |   ├── driver_routes.py
|   |   ├── schedule_routes.py
|   |   ├── dashboard_routes.py
|   |   ├── fuel_maintenance_routes.py
|   |   └── report_routes.py
|   |
|   ├── services/
|   └── schemas/
|
├── frontend/
|   ├── pages/
|   ├── css/
|   └── js/
|
├── tests/
|
├── requirements.txt
|
└── README.md
```

User Roles

Admin

- Manage Users
- Manage Routes
- Manage Vehicles
- Manage Drivers
- Assign Routes

Manager

- Manage Schedules
- Manage Fuel Records
- Manage Maintenance Records
- Generate Reports
- Monitor Dashboard

Driver

- View Assigned Routes
- View Schedule

User

- View Route Information
 - Track Bus Location
-

Testing Strategy

Black Box Testing

Authentication Testing

- Valid Login
- Invalid Login
- Logout

Route Testing

- Create Route
- Update Route
- Delete Route

Schedule Testing

- Conflict Detection
- Time Validation

Reporting Testing

- PDF Generation
 - Data Accuracy
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Deployment

Backend Deployment

- Render
- Railway
- VPS

Frontend Deployment

- Vercel
- Netlify

Database

- Firebase Realtime Database
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Future Enhancements

- SMS Notifications
 - Email Notifications
 - Mobile Application
 - AI-Based Route Optimization
 - Predictive Maintenance Analysis
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Team Contribution

Member	Module
Member 01	Authentication + Dashboard

Member	Module
Member 02	Route + Schedule
Member 03	Vehicle + Driver + Fuel & Maintenance + Reports

Git Workflow

Main Branch

Production Ready Code

Develop Branch

Integration Branch

Feature Branches

feature/authentication

feature/routes

feature/vehicles

feature/drivers

feature/schedules

feature/dashboard

feature/fuel-maintenance

feature/reports

Completion Checklist

- ☐ Requirements Analysis
- ☐ UML Diagrams
- ☐ Firebase Setup
- ☐ Authentication Module
- ☐ Route Module

- ☐ Vehicle Module
- ☐ Driver Module
- ☐ Schedule Module
- ☐ Dashboard Module
- ☐ Fuel Module
- ☐ Maintenance Module
- ☐ Reporting Module
- ☐ Testing
- ☐ Documentation
- ☐ GitHub Repository
- ☐ Final Submission